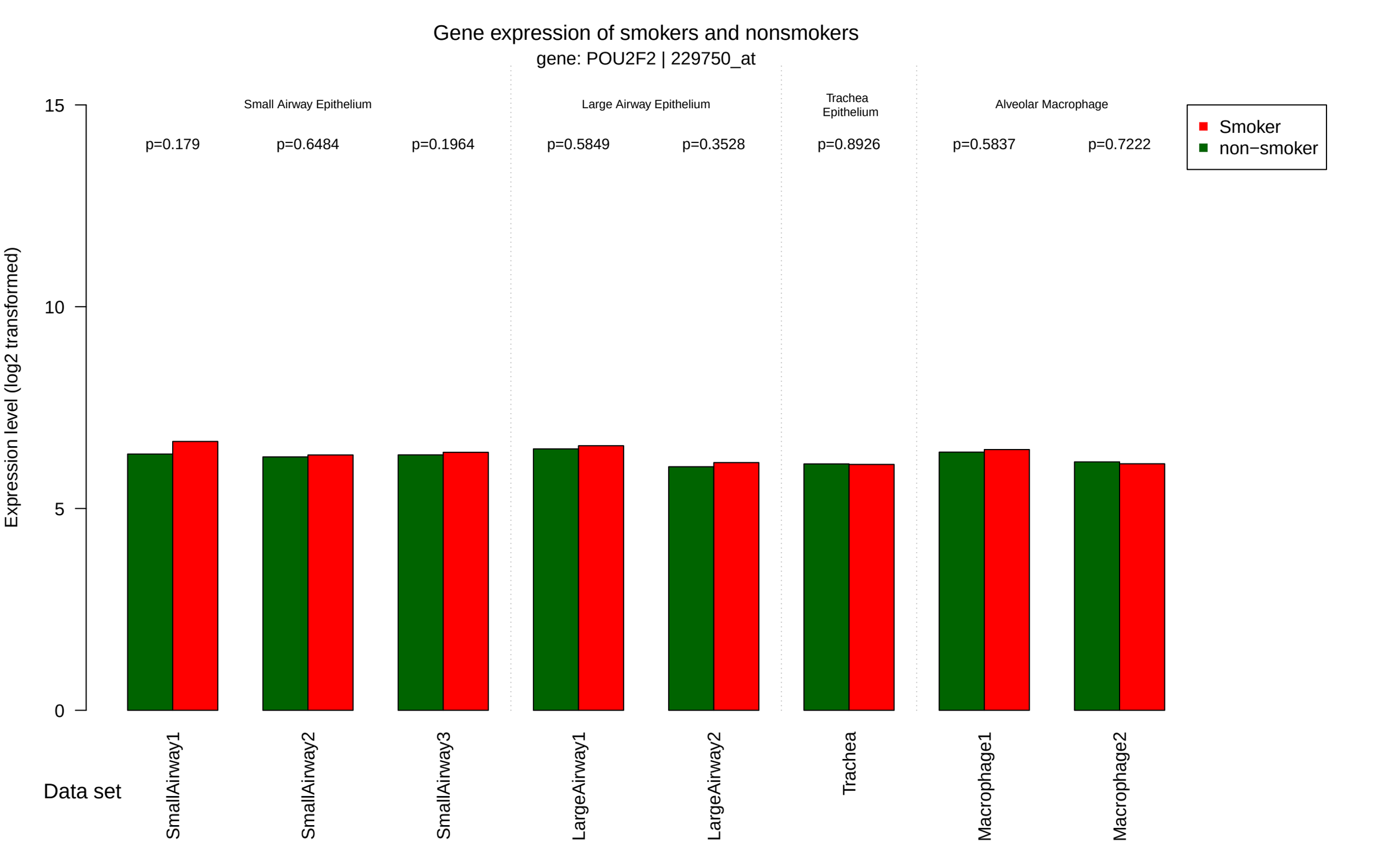
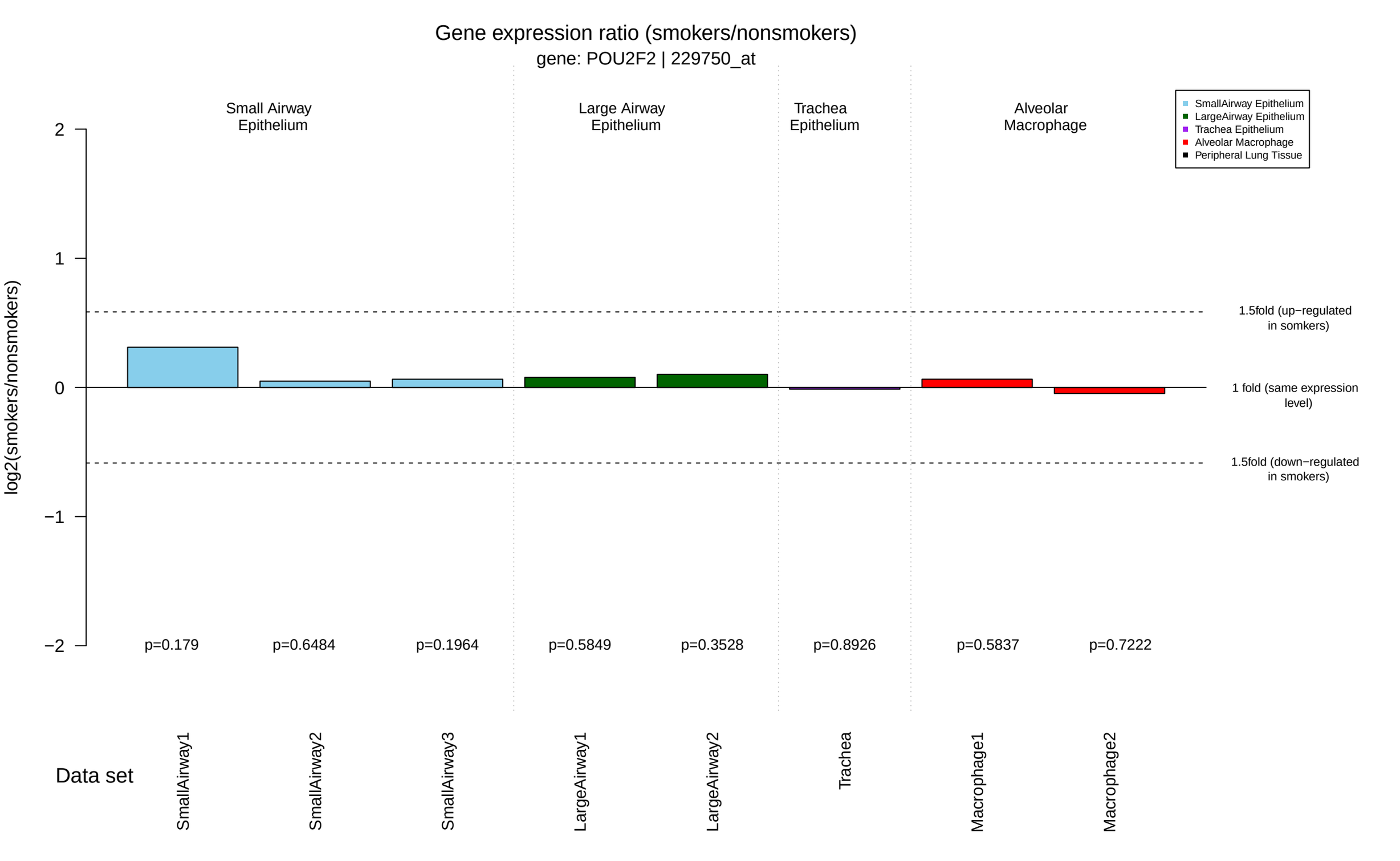
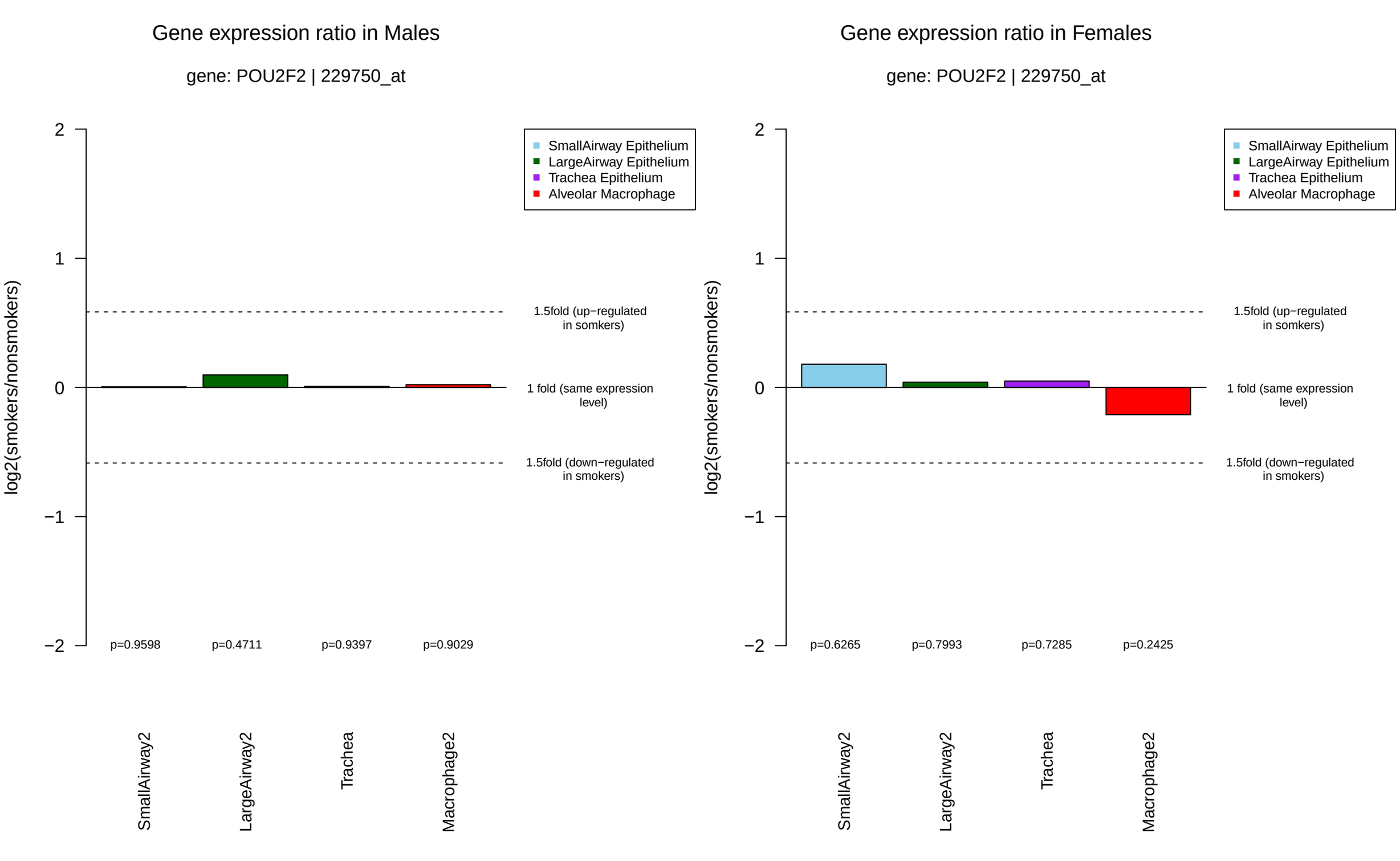
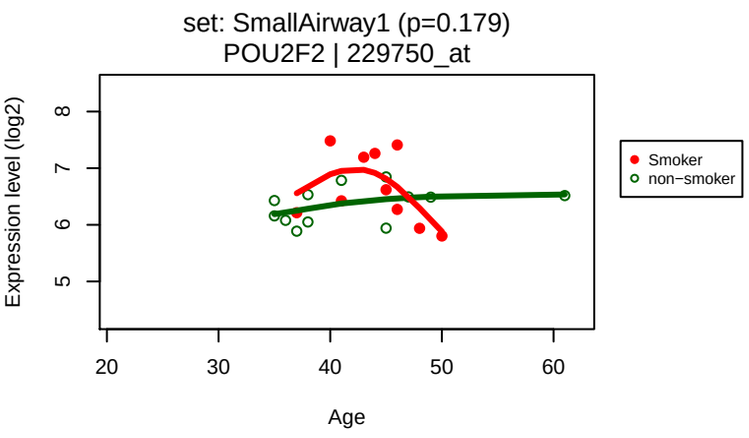
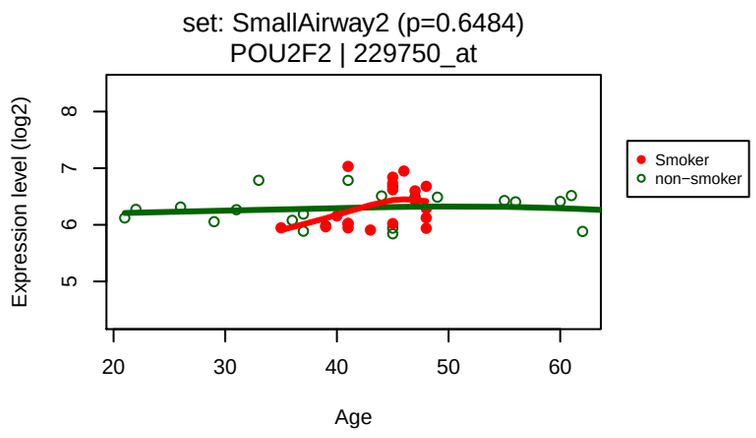
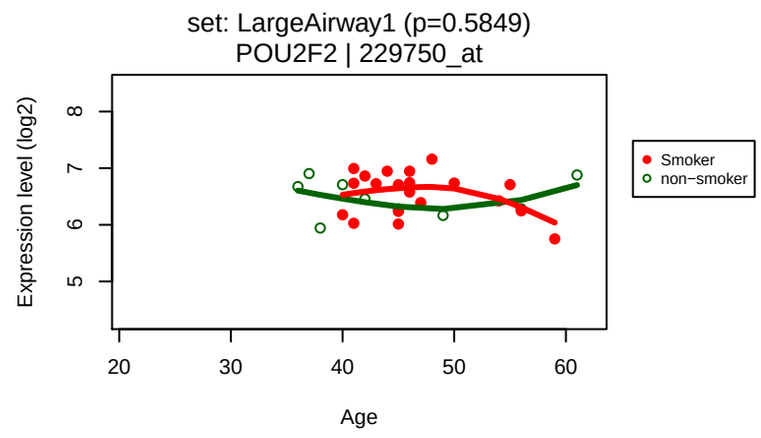

Gene expression of smokers and nonsmokers
gene: POU2F2 | 229750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.179
p=0.6484
p=0.1964
p=0.5849
p=0.3528
p=0.8926
p=0.5837
p=0.7222
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

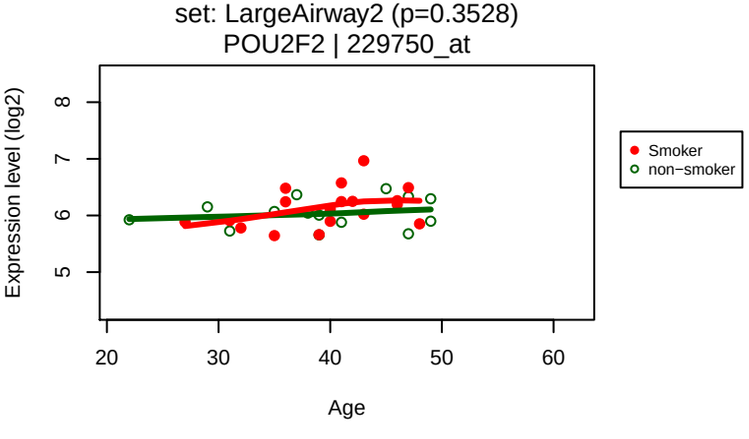

Gene expression ratio (smokers/nonsmokers)
gene: POU2F2 | 229750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.179
p=0.6484
p=0.1964
p=0.5849
p=0.3528
p=0.8926
p=0.5837
p=0.7222
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

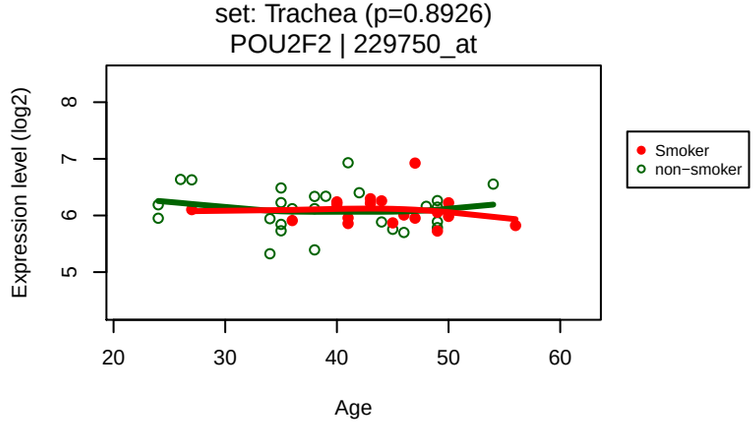

Gene expression ratio in Males
gene: POU2F2 | 229750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9598
p=0.4711
p=0.9397
p=0.9029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POU2F2 | 229750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6265
p=0.7993
p=0.7285
p=0.2425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

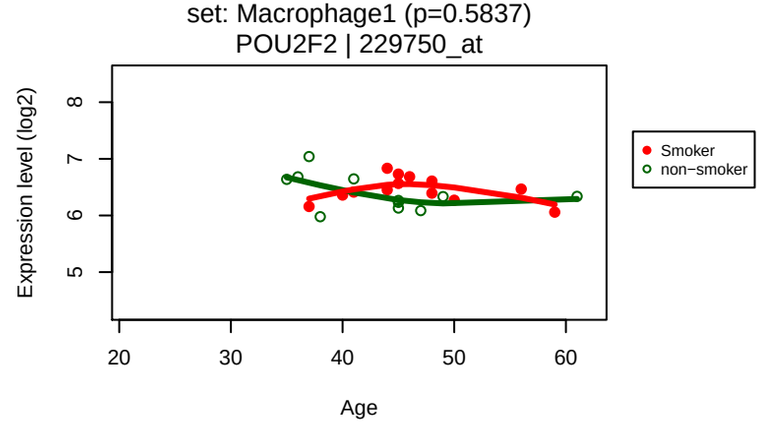

set: SmallAirway1 (p=0.179)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker

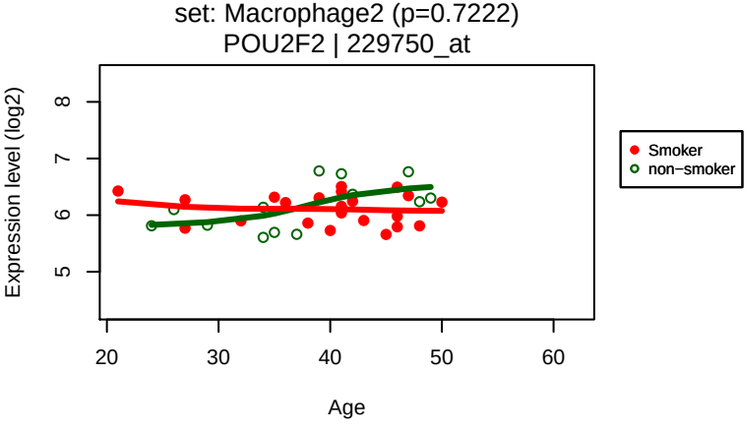

set: SmallAirway2 (p=0.6484)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker

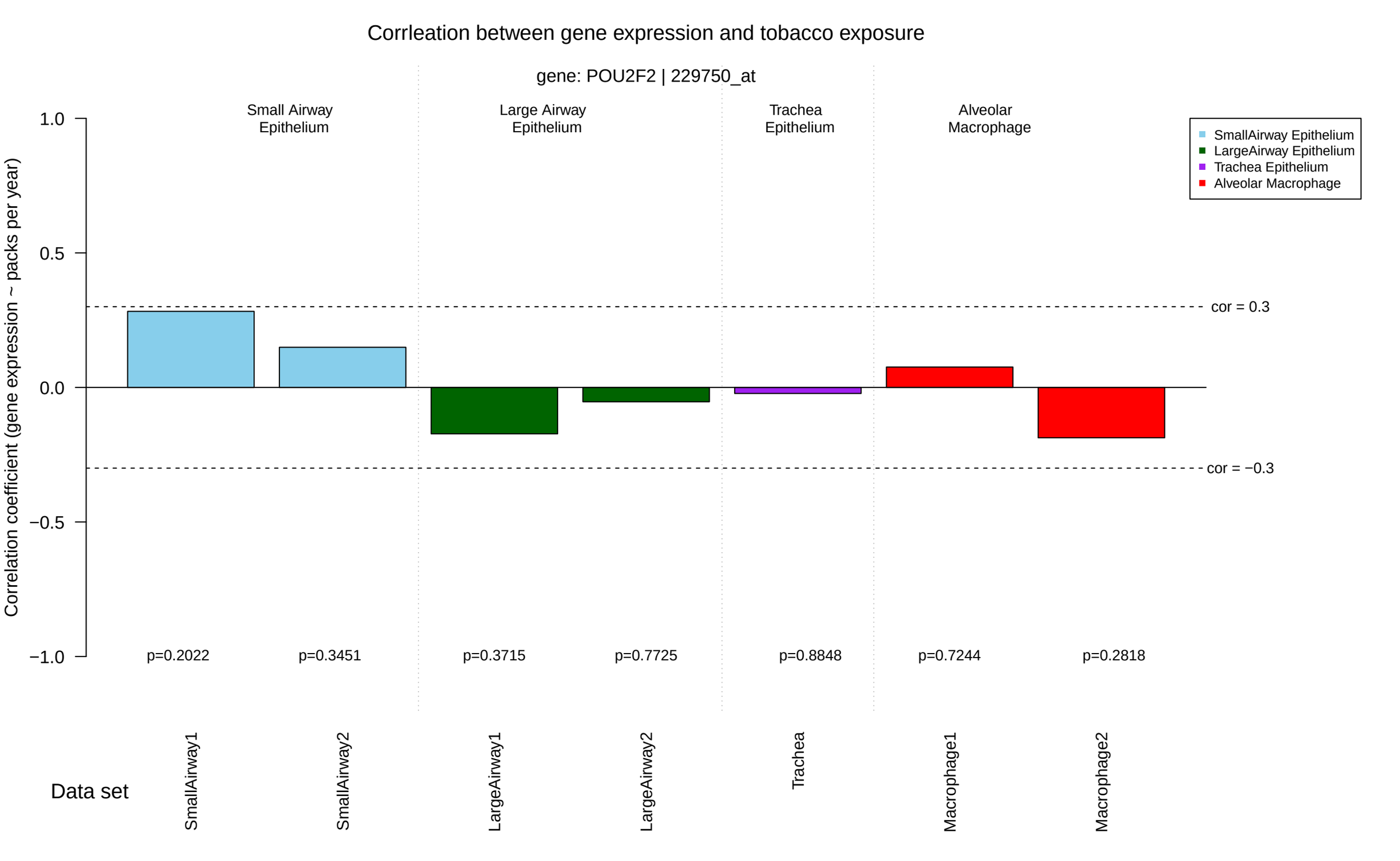

set: LargeAirway1 (p=0.5849)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3528)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8926)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5837)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7222)
POU2F2 | 229750_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POU2F2 | 229750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2022
p=0.3451
p=0.3715
p=0.7725
p=0.8848
p=0.7244
p=0.2818
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2